AMARNATH S. PATEL

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Systems and scientific software — C++23/CUDA, Rust, Python. Hardware to GPU kernels to shipped desktop apps.

EDUCATION

University of Central Florida — **Undergraduate Student** — **4.00 GPA**
B.S. Physics (Optics & Lasers) and Mathematics, Computer Science minor — August 2025 – Expected May 2029

- **Computer science:** Data Structures & Algorithms, Computer Architecture, Computer Logic Design, Object-Oriented Programming, Discrete Structures, C Programming, Deep Learning
- **Mathematics:** Linear Algebra (proof-based, in progress), Complex Analysis, Partial Differential Equations (in progress), Applied Linear Algebra, Ordinary Differential Equations, Honors Calculus III

Florida Atlantic University — **University coursework via FAU High School, ages 14–18** — **3.66 GPA**
111 credit hours completed between ages 14 and 18, alongside high school — August 2021 – May 2025

EXPERIENCE

Undergraduate Researcher, Instrumentation Software — UCF Astrophotonics Lab, CREOL — Aug 2025 – Present

- Own the lab's acquisition software: a Python stack that drives four instruments as one system — tunable IR laser, InGaAs camera, fiber switch and motorized polarization controller — across three transports (GPIB/VISA, RS-232, GigE Vision), so a full all-port × C-band measurement sweep runs unattended. Days of manual bench operation became a single hands-off run.
- Closed-loop control and fault handling in the acquisition path: polarization optimized in-loop against a live fringe-contrast objective, saturated frames detected and rejected as they arrive, so an overnight run does not silently produce unusable data.
- Built the numerical pipeline that turns raw frames into results — FFT sideband isolation and demodulation, Butterworth filtering, and a joint nonlinear optimization over three coupled parameters (NumPy/SciPy) — validated at ≈ 98% reconstruction fidelity against simulated ground truth.
- Own the OS and node software for PolyOculus, a networked array of 8 telescopes each driven by a Jetson Nano. Brought up mainline Linux 7.1 and Arch Linux ARM to replace NVIDIA's end-of-life Ubuntu 18.04 image: traced boot failures to an SD-card regulator behind a modular Tegra I2C driver, U-Boot's 14 MiB initramfs load window, and missing Tegra PWM/host1x drivers, and built a kernel that boots with no initramfs. Any node rebuilds from the monorepo in about 15 minutes (Ansible, canary-first fleet upgrades).
- Wrote the node control loops: autoguiding from a USB3 guide camera (driven through its vendor UVC extension unit for 1 µs exposure control) to INDI mount guide pulses, cutting simulated tracking error from 3.01″ to 0.29″ RMS, and a PI-loop piezo steering mirror that holds a laser spot to 0.2 px.

Undergraduate Research Assistant — UCF Physics Department — Mar 2026 – Present
Physics Education Research — Dr. Zhongzhou Chen (NSF Award 2421299; Gates Foundation INV-076932)

- Built ESTELA (Rust, Tauri 2), a desktop application that generates multi-version isomorphic exams from a structured bank of 615 problems across 29 banks, 13 topic areas and 11 question types, with auto-generated answer keys and export to four formats. Shipped to external adopters outside the university following a faculty workshop.
- Built an agentic generation–evaluation pipeline over the Azure OpenAI batch API that drafts candidate problems and validates them against instructor criteria before a human ever reviews them.

SELECTED SOFTWARE

Stardust — finding new near-Earth asteroids in archival sky-survey images. Python, PyTorch, CUDA — Oct 2026

- Knight Hacks IX team project; I built the detection and orbit pipeline: orbit fit from a new asteroid's first 4 days, a batched N-body integrator in PyTorch on the GPU (100,000 orbit clones × 1 year in 23 s on a laptop RTX 4070), search regions projected onto years of archival ZTF images, a U-Net detector and a classic streak filter, and track linking gated by an orbit refit.
- Packaged for SLURM GPU clusters (job arrays, an Apptainer image with pinned dependencies), with cached and rate-limited downloads from the public archives, live target discovery into Databricks Delta tables with MLflow, and the model and recovered dataset published to Hugging Face. Recovered 167 verified, never-connected sightings of 22 once-risky asteroids in a blind 72-object backtest.

CELERIS — electromagnetic solver and metalens design pipeline. C++23, CUDA, Python, CMake — 2026

- Wrote a Maxwell solver from scratch rather than wrapping an existing kernel, then built the design and analysis tooling on top: parameter sweeps, gradient-based (Adam) inverse design, Monte-Carlo tolerance analysis, and fabrication-ready GDSII export. Python bindings and a native Qt desktop GUI.
- Correctness is enforced, not assumed: a validation suite checks every layer against closed-form solutions, an independent method, or energy conservation (agreement from 1e-6 to 1e-15), and cross-checks the whole solver against two third-party implementations to ≈ 1e-7.
- Implemented Rayleigh–Sommerfeld propagation as a CUDA kernel and benchmarked it against the project's own optimized 16-core CPU path: 4.8–5.8× at 92k elements, agreeing to machine precision, with automatic CPU fallback. Diagnosed the kernel as memory-bound rather than compute-bound from the way the speedup declined as the grid grew. *Sole-author manuscript in preparation.*

Systems and tooling — C, C++, Rust, Go — 2023 – Present

- hush, an interactive shell in C; limebar, a featherweight Wayland status bar in C; constrict, a build system in Go written to replace make; coat, a universal theme applier in Rust; hazmat, a small password manager in C99. Tools I use daily on my own machine, not demos.
- slashbinslash.sh, temporary file sharing and storage; apatel.co, a hand-written static site with no framework, no build step and no third-party requests at runtime.

TECHNICAL SKILLS

Languages: C, C++ (C++23), Rust, Python, Go, JavaScript/TypeScript, CUDA, VHDL, Shell (Fish, Bash)

Frameworks and libraries: Qt/PySide6, Tauri 2, Nuxt/Vue, Express, Electron, Three.js, NumPy, SciPy

Systems and tools: Linux (Arch, daily driver), Git, CMake, Docker, WireGuard, Wayland, embedded Linux (mainline kernel and U-Boot on NVIDIA Jetson; Raspberry Pi), Ansible, INDI, UVC/V4L2, GPU/CUDA profiling, GPIB/VISA, RS-232, GigE Vision, LaTeX/Typst

HONORS

Florida Bright Futures — Florida Academic Scholars (highest tier; 100% tuition) — 2025
Lockheed Martin Award, "Highest Level of Engineering Excellence," AEV Competition — 2024